

Arrays in JavaScript

An **array** is a collection of items stored in a single variable. It lets you store multiple values — like a list of names, numbers, or even other arrays.

1. Creating Arrays

```
let fruits = ["apple", "banana", "mango"];  
let numbers = [10, 20, 30, 40];
```

You can mix data types, but it's better to keep arrays consistent.

```
let mixed = [1, "hello", true];
```

2. Accessing and Modifying Elements

Arrays use **zero-based indexing**.

```
let fruits = ["apple", "banana", "mango"];  
console.log(fruits[0]); // "apple"  
fruits[1] = "orange";    // change "banana" to "orange"  
console.log(fruits);     // ["apple", "orange", "mango"]
```

3. Array Length

```
let colors = ["red", "green", "blue"];  
console.log(colors.length); // 3
```

4. Common Array Methods

a. **push()** – Add item at the end

```
colors.push("yellow");
```

b. **pop()** – Remove last item

```
colors.pop();
```

c. **shift()** – Remove first item

```
colors.shift();
```

d. **unshift()** – Add item at the beginning

```
colors.unshift("pink");
```

5. Looping Through Arrays

Using `for` loop:

```
let fruits = ["apple", "banana", "mango"];
for (let i = 0; i < fruits.length; i++) {
  console.log(fruits[i]);
}
```

Using `for...of` loop:

```
for (let fruit of fruits) {
  console.log(fruit);
}
```

6. `map()` – Transform Array

Creates a new array by applying a function to each item.

```
let numbers = [1, 2, 3];
let doubled = numbers.map(num => num * 2);
console.log(doubled); // [2, 4, 6]
```

7. `filter()` – Filter Items

Returns a new array of items that match a condition.

```
let scores = [30, 50, 90, 20];
let passed = scores.filter(score => score >= 50);
console.log(passed); // [50, 90]
```

8. Other Useful Methods

Method	Description
<code>includes()</code>	Checks if an item exists in the array
<code>indexOf()</code>	Finds index of an item
<code>slice()</code>	Extracts part of the array
<code>join()</code>	Joins array into a string

Example:

```
let names = ["Ali", "Sara", "John"];
console.log(names.includes("Sara")); // true
console.log(names.indexOf("John")); // 2
console.log(names.join(" - "));      // Ali - Sara - John
```

Summary

- Arrays store lists of data in one variable.
- Use indexing to access or update items.
- Common methods like `push`, `pop`, `map`, `filter` make array handling easier.
- Looping is essential for displaying or processing lists.